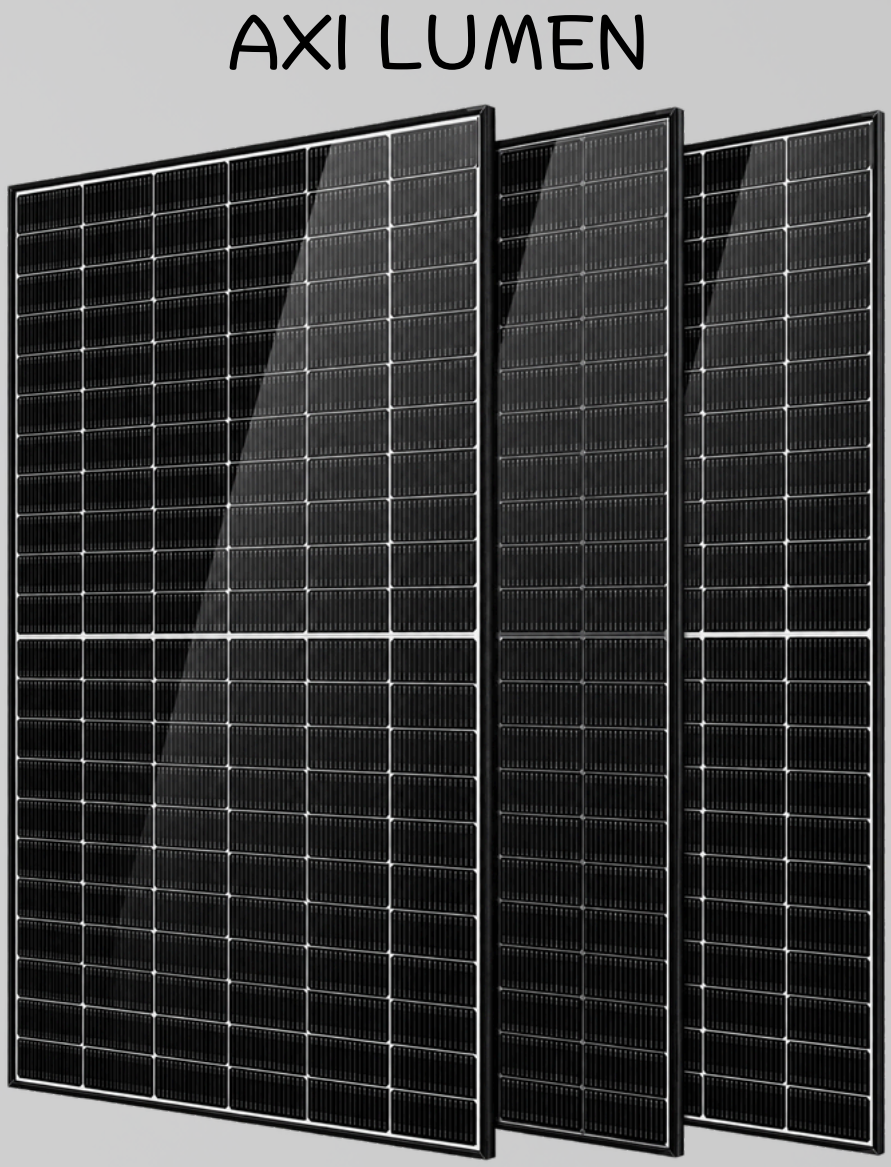
# 620~640 Watt

# AXI LUMEN 620-640N-DG BF




## AXI PREMIUM SOLAR PANELS

**SMBB Half-Cell Technology**
More uniform current collection capability, reducing the current heat loss of the internal cells.

**Higher Output Power**
The output power of 132 half-cells monocrystalline modules is up to 640W.

**Low Light Features**
Higher performance under low light environment.

**Harsh Environmental Adaptability**
Strict salt spray and ammonia corrosiontest by the third party.

**N-type With Very Low LID**
N-type solar cell has very low LID naturally which can reduce power degradation.

**Dual-Sided Power Generation**
Gain increases with backside exposure to light, significantly reducing LCOE.

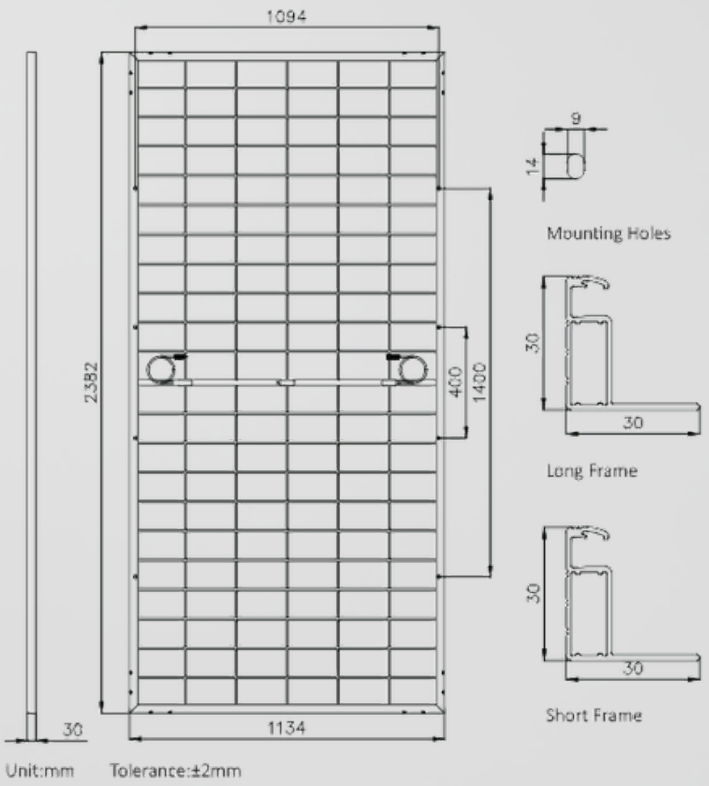
**UL 61730**
**IEC61215:2021**
**IEC61730:2023**
**ISO9001:2015:** ISO Quality Management System
**ISO14001: 2015:** ISO Environment Management System
**ISO45001: 2018:** Occupational Health and Safety
**IEC62941:** Guideline for module design qualification and type approval

15 15-year product warranty

30 30-year linear power output warranty

# AXI LUMEN 620-640N-DG BF

## MECHANICAL DIAGRAMS



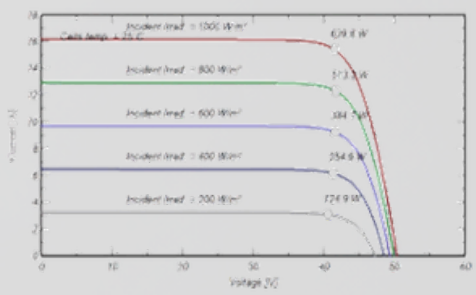

Unit:mm Tolerance:±2mm

## MECHANICAL PARAMETERS

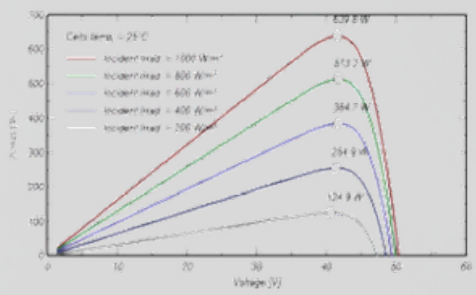
| | |
|---|---|
| Weight | 33.7kg |
| Dimension | 2382×1134×30mm |
| Cell Orientation | 132(6×22) |
| Junction Box | IP68, three diodes |
| Output Cable | 4mm²,±300mm(length can be customized) |
| Connector | MC4 compatible |
| Glass | 2.0+2.0mm AR coated heat strengthened glass |
| Frame | Anodized aluminum alloy frame |
| Packaging | 36pcs per pallet/720pcs per 40'HC |

## CURVES OF PV MODULE

AX640TC-MHC-D



AX640TC-MHC-D



## ELECTRICAL CHARACTERISTICS

| Module Type | AX620TC-MHC-D | | AX625TC-MHC-D | | AX630TC-MHC-D | | AX635TC-MHC-D | | AX640TC-MHC-D | |
|---|---|---|---|---|---|---|---|---|---|---|
| Testing Condition | STC | NMOT | STC | NMOT | STC | NMOT | STC | NMOT | STC | NMOT |
| Maximum Power(Pmax/W) | 620 | 484 | 625 | 488 | 630 | 491 | 635 | 495 | 640 | 499 |
| Open Circuit Voltage(Voc/V) | 49.40 | 49.05 | 49.63 | 49.29 | 49.87 | 49.52 | 50.11 | 49.76 | 50.35 | 50.00 |
| Short Circuit Current(Isc/A) | 15.95 | 12.44 | 16.00 | 12.48 | 16.05 | 12.52 | 16.10 | 12.56 | 16.15 | 12.60 |
| Maximum Power Voltage(Vmp/V) | 41.05 | 41.50 | 41.25 | 41.70 | 41.45 | 41.90 | 41.65 | 42.10 | 41.84 | 42.30 |
| Maximum Power Current(Imp/A) | 15.10 | 11.65 | 15.15 | 11.69 | 15.20 | 11.73 | 15.25 | 11.76 | 15.29 | 11.80 |
| Module Efficiency(%) | 22.9 | | 23.1 | | 23.3 | | 23.5 | | 23.7 | |

* Under Standard Test Conditions (STC) of irradiance of 1000 W/m², spectrum AM 1.5 and cell temperature of 25°C.

* Under Nominal Module Operating Temperature (NMOT), irradiance of 800 W/m², spectrum AM 1.5, ambient temperature 20°C, wind speed 1 m/s.

## ELECTRICAL CHARACTERISTICS WITH 10% SOLAR IRRADIATION RATIO (BNPI)

| | | | | | |
|---|---|---|---|---|---|
| Maximum Power(Pmax/W) | 682 | 688 | 693 | 699 | 704 |
| Open Circuit Voltage(Voc/V) | 49.40 | 49.63 | 49.87 | 50.11 | 50.35 |
| Short Circuit Current(Isc/A) | 17.55 | 17.60 | 17.66 | 17.71 | 17.77 |
| Maximum Power Voltage(Vmp/V) | 41.05 | 41.25 | 41.45 | 41.65 | 41.84 |
| Maximum Power Current(Imp/A) | 16.61 | 16.67 | 16.72 | 16.77 | 16.82 |

*Rear side power gain:The additional gain from the rear side compared to the power of the front side at the standard test condition.It depends on mounting (structure,height,tilt angle etc.)and albedo of the ground.

## OPERATING PARAMETERS

| | |
|---|---|
| Operational Temperature | -40°C~+85°C |
| Power Output Tolerance | 0~3% |
| Maximum System Voltage | 1500V |
| Maximum Series Fuse Rating | 30A |
| Nominal Operating Cell Temperature | 45±2°C |
| Protection Class | Class II |
| Bifaciality | 80±5% |
| Fire Rating | IEC Class A |

## MECHANICAL LOADING

| | |
|---|---|
| Front Side Maximum Static Loading | 5400Pa |
| Rear Side Maximum Static Loading | 2400Pa |
| Hailstone Test | 25mm Hailstone at the speed of 23m/s |

## TEMPERATURE RATINGS (STC)

| | |
|---|---|
| Temperature Coefficient of Isc | +0.04%/°C |
| Temperature Coefficient of Voc | -0.23%/°C |
| Temperature Coefficient of Pmax | -0.28%/°C |

*The actual test value may be slightly deviated from the technical parameters due to the difference in test methods.

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Note: Please read the safety and installation manual before using the product.
We reserve the right of final interpretation.
The specifications in this datasheet are subject to change without notice.

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